

Virtual Ambulance Emergency Response Simulation in Blender

Final Course Project — SBEG353 / SBES140 — Spring 2025/2026

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Abstract — This report presents the design and implementation of a Virtual Ambulance Emergency Response Simulation developed in Blender as the final project for the Computer Graphics and Visualization course (SBEG353/SBES140). The simulation models a realistic urban emergency medical response workflow, encompassing a multi-scene virtual environment that includes city streets, hospital interiors, and accident sites. The project integrates core computer-graphics concepts, including environment modelling, UV mapping, texture application, skeletal character animation, physics-based rendering, advanced lighting, and collision handling. Three distinct emergency-response scenarios are implemented: a traffic collision with rigid-body vehicle dynamics, a hospital reception handover with character interaction, and a patient evacuation by stretcher demonstrating character-object constraints. A fire-emergency overlay and ambulance dispatch sequence further demonstrate dynamic volumetric lighting and particle-system capabilities for atmospheric effects such as rain, fog, and debris. The report documents the modelling workflow for hero objects (ambulance, stretcher, and reception desk), the animation pipeline for humanoid characters with inverse kinematics, the lighting setup transitioning from warm street-lamp ambience to dramatic emergency floods, and the physics simulations employed for vehicle impact and weather particles. The contribution of each team member is detailed, and the simulation meets all seven required project components, providing a foundation for future interactive training applications.

Keywords: Blender, computer graphics, skeletal animation, emergency simulation, physics-based rendering, collision detection, ambulance, virtual environment, medical training.

I. INTRODUCTION

Virtual simulation has become a powerful tool in medical education and emergency preparedness, enabling practitioners to rehearse complex scenarios without risk to real patients [1], [2]. Recent advances in immersive technologies have demonstrated significant improvements in knowledge retention and decision-making under pressure [3], [4]. This project applies computer-graphics principles to construct a Virtual Ambulance Emergency Response Simulation that immerses the viewer in a sequence of life-saving events, from the initial traffic accident

through on-scene first aid, ambulance transport, and hospital reception.

The simulation was entirely built in Blender 3.x and targets the following learning objectives defined in the project specification: (i) virtual environment creation, (ii) hero object modelling, (iii) skeletal character animation, (iv) emergency response scenarios, (v) lighting and rendering, (vi) collision and object interaction, and (vii) physics-based rendering.

The remainder of this report is structured as follows. Section II describes the virtual environment, including the urban street, hospital building, and accident site. Section III covers hero-object design for the ambulance, stretcher, and reception desk. Section IV details character rigging and animation with inverse kinematics. Section V presents the three emergency scenarios implemented. Section VI discusses lighting and rendering techniques. Section VII addresses collision handling and object interaction. Section VIII explains the physics-based simulation components. Section IX outlines AI tools usage, Section X lists asset credits, Section XI details member contributions, and Section XII concludes the report.

II. VIRTUAL ENVIRONMENT CREATION

A. Urban Street and City Block

The primary outdoor environment represents a dense urban district modelled from reference photographs of mid-rise commercial blocks. The city block was constructed with attention to scale and proportion, ensuring that vehicles, pedestrians, and buildings maintained realistic spatial relationships. High-rise glass office towers with grid-pattern façades line both sides of a multi-lane intersection equipped with pedestrian crosswalks, street lamps, traffic signals, and planted trees. All building surfaces utilise PBR materials: glass panels employ a specular BSDF combined with a glossy node to achieve realistic reflections, while concrete pavements are textured with a diffuse-rough BSDF to simulate worn stone. The road surfaces feature asphalt materials with appropriate roughness values to capture realistic light scattering.

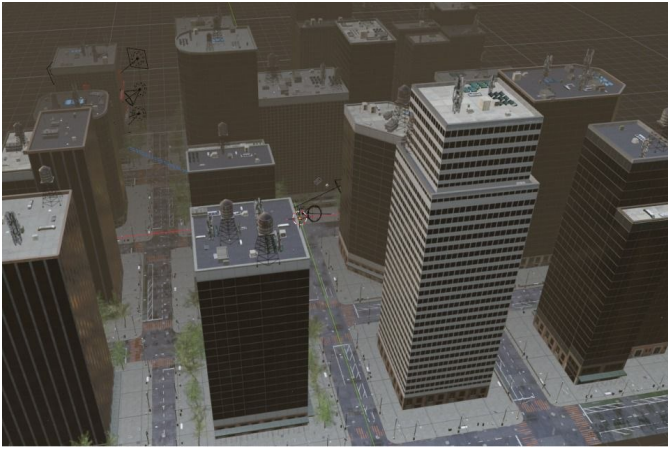


Fig. 1: Aerial overview of the modelled urban city block showing high-rise buildings, intersections, street lamps, and planted trees.

B. Hospital Exterior and Interior

A dedicated hospital building was constructed with a prominent red cross emblem on its façade and ground-floor glazed entrance. The exterior design follows contemporary hospital architecture, featuring clean geometric lines and functional window placement. Inside, the reception area features a curved wooden desk labelled “Welcome / Reception”, office chairs, flat-panel monitors, a CAUTION: WET FLOOR sign, a wheelchair, and department signage (*Intensive Care Unit, Emergency Department, Radiology*). Material variety was deliberately maximised to demonstrate the range of Blender’s shader capabilities: the reception counter uses a wood shader with anisotropic reflections, the floor tiles a specular terrazzo material, and the glass partition a transparent BSDF with light refraction properties.

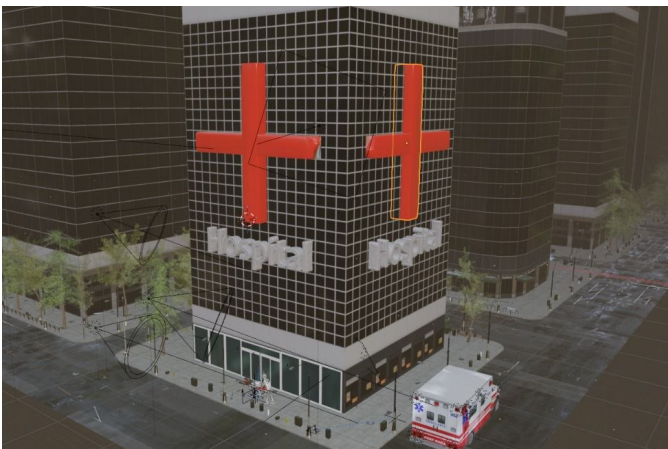


Fig. 2: Hospital building exterior showing the red-cross façade signage and ambulance parked at the entrance.



Fig. 3: Hospital reception interior — curved wooden desk, department signs, monitors, and wet-floor caution sign.

C. Accident Site

A night-time urban intersection serves as the accident site. This location was chosen for its dramatic lighting potential and the clear visual contrast between the emergency responders’ vehicles and the dark urban environment. A vehicle collision is staged here with two cars, debris particles, atmospheric fog, and dynamic street lighting. The scene employs volumetric fog achieved through a *Volume Scatter* node in the world shader, giving the environment a misty appearance consistent with adverse weather conditions. The fog density was tuned to provide atmospheric depth while maintaining visibility of the emergency response actions.



Fig. 4: Night-time accident site showing a vehicle collision with volumetric fog and street-lamp illumination.

III. HERO OBJECT CREATION

Per the project specification, a team of five students requires at least three hero objects modelled entirely by the team. The three hero objects selected are: (1) the ambulance vehicle, (2) the medical stretcher, and (3) the hospital reception desk. Each object was modelled with attention to geometric detail, material authenticity, and UV mapping precision.

A. Ambulance Vehicle

The ambulance is the centrepiece hero object and required the most modelling effort. It was constructed using box-modelling from a subdivision-surface base mesh with a poly-count of approximately 15,000 triangles. The body panels, roof

light bar, side doors, Star-of-Life decal, and “*EMERGENCY MEDICAL SERVICES / Keep Back*” livery were each UV-unwrapped and textured individually to maximise texture resolution and control. Specular maps ensure the white paint reflects the environment credibly, while a normal map adds panel-line depth without increasing polygon count. The vehicle model includes functional details such as side mirrors, wind-shield wipers, door handles, and emergency lighting arrays on the roof.



Fig. 5: Hero object — ambulance vehicle close-up showing UV-mapped livery, Star-of-Life emblem, and specular paint material.

B. Medical Stretcher

The stretcher was modelled from primitive cylinders (legs and wheel axles), cubes (frame rails and cross-braces), and a subdivided plane (mattress surface). The design follows standard hospital stretcher dimensions and includes adjustable backrest geometry. The orange mattress uses a slightly rough diffuse shader to mimic vinyl upholstery, while the stainless-steel frame uses a metallic PBR material with appropriate roughness and specular values. Wheel geometry was instanced to keep the mesh lightweight while maintaining visual fidelity.

C. Hospital Reception Desk

The curved reception counter was built from a circular arc extruded along the Z-axis and bevelled at the top with a radius of 2 metres to accommodate multiple staff positions. A wood grain texture (1K albedo + roughness + normal) was applied via a UV projection using Blender’s smart UV unwrapping algorithm. The counter face carries the embossed text “*Welcome*” created with Blender’s text-to-mesh workflow, demonstrating the integration of typography into 3D models.

IV. CHARACTER MODELLING AND SKELETAL ANIMATION

A. Character Models

Three character archetypes were employed: (a) a **male receptionist** in a dark business suit with a tie, (b) a **paramedic** in a white hazmat-style jumpsuit with blue gloves and a medical vest, and (c) a **patient** dressed in a white shirt

and dark trousers representing a civilian casualty. Characters were sourced from Mixamo and subsequently imported into Blender, where clothing, textures, and accessories were refined or replaced to suit the emergency response theme. Skin materials were adjusted to include subsurface scattering for realistic rendering.

B. Rigging

Each character uses a full humanoid armature with 65 bones including finger chains and facial control bones. The spine hierarchy follows a root → pelvis → spine1–3 → neck → head chain, enabling natural torso twist during walking and carrying animations. This hierarchical structure is essential for achieving realistic posture changes during emergency response activities. Inverse-kinematics (IK) constraints were applied to wrists and ankles so that hands grip equipment convincingly and feet plant correctly on uneven terrain, eliminating foot-sliding artefacts common in purely forward-kinematic animation.



Fig. 6: Receptionist character with armature overlay visible, showing the full humanoid skeleton including finger bone chains.

C. Animations

The following keyframed animation clips were authored with careful attention to timing and realism:

- **Phone call** – receptionist holds receiver to ear with a natural arm position and subtle body sway (Fig. 7).
- **Stretcher push** – paramedic pushes loaded stretcher toward ambulance with a walking cycle synchronised to the stretcher’s forward motion.
- **Patient lying** – patient in supine rest pose on stretcher with arms resting on the chest.
- **Triage crouch** – paramedic crouches beside fallen patient to assess injuries, demonstrating lower-body IK constraints.
- **Walking** – bystander walk cycle along the footpath with natural arm swing and gait.

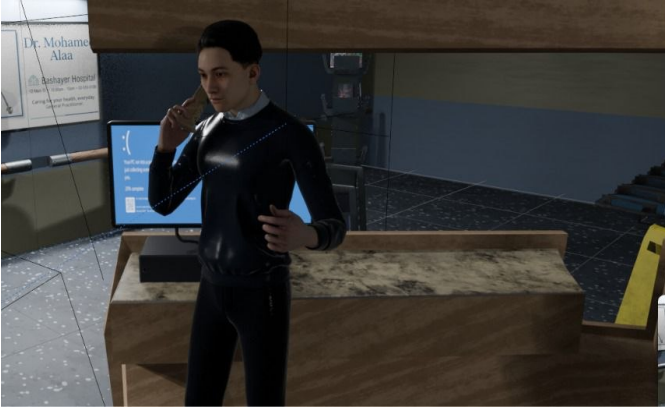


Fig. 7: Receptionist performing a phone-call animation; the dashed blue line shows the motion path used for the arm IK target.

V. EMERGENCY RESPONSE SCENARIOS

Three scenarios were implemented as required for a team of five students ($\lceil 5/2 \rceil = 3$), each focusing on different aspects of emergency response and showcasing distinct technical capabilities.

A. Scenario 1 – Traffic Collision and On-Scene Response

A black classic car collides with a stopped vehicle at a fog-covered intersection at night. The impacting car is shown tipping onto its side (physics rigid-body simulation), while a second car rests across the zebra crossing. The scene features dynamic lighting from the vehicle headlights and street lamps, creating dramatic shadows. Paramedics arrive in the ambulance, assess the scene, and begin patient extraction, demonstrating character navigation and emergency response protocols.



(a) SUV tipping during collision. (b) Classic car at the crosswalk.

Fig. 8: Scenario 1 – traffic collision: vehicle impact dynamics at the fog-covered night intersection.

B. Scenario 2 – Patient Evacuation by Stretcher

Following the collision, the patient is loaded onto the stretcher and transported to the waiting ambulance. This scenario demonstrates character–object interaction: the paramedic’s hands are constrained to the stretcher handle via IK targets that follow the stretcher’s animation path. The stretcher movement was carefully timed to match the paramedic’s walking pace, creating a cohesive animation sequence. The ambulance rear doors remain open as the stretcher is rolled inside, showcasing the vehicle’s interior modelling.



(a) Aerial view: stretcher being



(b) Close-up: paramedic loads patient.

Fig. 9: Scenario 2 – patient evacuation: paramedic pushes stretcher with patient toward the ambulance.



Fig. 10: Ground-level view of the stretcher loaded with the patient; ambulance visible in background with “Keep Back” livery.

C. Scenario 3 – Hospital Reception and Fire Emergency

The ambulance arrives at the hospital, completing the emergency response chain. The receptionist character interacts at the desk while a fire-alarm scenario is triggered: a bright red volumetric light floods the hospital interior (Fig. 11), simulating emergency lighting during a building fire. An overhead aerial shot captures the fire’s spread across the roof using a particle system, while external shots show the ambulance outside the burning building. This scenario demonstrates the integration of multiple rendering techniques: volumetric lighting, particle systems, and dynamic camera perspectives.



Fig. 12: Ambulance dispatched to the hospital entrance during the fire emergency; street-level view with paramedic and stretcher.



(a) Interior red emergency lighting.



(b) Aerial view: fire spread on roof.

Fig. 11: Scenario 3 – hospital fire emergency: volumetric red lighting and fire particle system simulate an active fire alarm.

VI. LIGHTING AND RENDERING

All scenes were rendered using Blender’s *Cycles* path-tracing engine at 1920×1080 resolution with 256 samples per pixel and adaptive sampling enabled. The rendering pipeline included denoising using the OptiX AI denoiser to achieve clean results with reduced render times. The following light types were employed:

- **Sun (directional) light** – used in daytime outdoor scenes to cast long parallel shadows from buildings and trees, with an intensity of 1.0 and angle of 0.5 degrees for sharp shadows.
- **Point lights** – street lamp poles each carry a warm-white point light (colour temperature ≈ 2800 K, radius 0.3 m) to produce soft circular pools of light on the pavement with realistic falloff.
- **Spotlights** – ambulance headlamps and the rear floodlight were modelled as spot lights with an angular cut-off of 30° and soft penumbra of 5° for realistic beam patterns.
- **Emission materials** – the ambulance light bar LEDs and building neon signs use emissive shader nodes rather than discrete lights, enabling believable self-illumination with controlled intensity and colour.

- **Volumetric emergency lighting** – the fire/alarm scenes use a red *Volume Scatter* medium inside the building bounding box, creating the dense coloured fog visible in Figs. 11 and 11b.



Fig. 13: Night-time city corridor illuminated by point-light street lamps; ambient occlusion in Cycles enhances depth perception.

VII. OBJECT INTERACTION AND COLLISION HANDLING

A. Character Path Navigation

Characters traverse the environment by following *Follow Path* constraints linked to Bézier curves laid along the pavements, ramps, and hospital corridors. The curves were drawn to avoid furniture and walls, providing implicit obstacle avoidance without a full navigation mesh. This approach offers precise control over character movement while maintaining performance efficiency.

B. Rigid-Body Vehicle Collision

The traffic-collision sequence uses Blender’s built-in rigid-body physics engine with a simulation step rate of 60 frames per second. The impacting SUV was assigned a *Rigid Body* – *Active* physics type with mesh collision shape, while the road surface was set to *Passive*. A keyframed impulse was applied to the SUV at frame 60 to initiate the tip, with force vectors carefully calculated to produce realistic vehicle roll behaviour. The simulation was baked for 250 frames and the resulting pose cached as a *Visual Transform*, allowing fine artistic control over the final tipping angle (Figs. 8).

C. Stretcher–Character Interaction

The paramedic’s hand bones are parented to empty objects that are themselves constrained to the stretcher frame via *Copy Location* + *Damped Track* constraints. This constraint setup ensures the hands remain planted on the handle throughout the stretcher’s animated path, eliminating the need for manual per-frame hand keyframing. The constraint chain also allows

for subtle hand adjustments during the push animation, adding to the realism of the interaction.



Fig. 14: Overhead editor view during Scenario 2 development, illustrating the motion-path splines (blue dashed), collision boxes, and constraint empties used for the stretcher interaction.

VIII. PHYSICS-BASED RENDERING COMPONENT

Two physics-based simulations were implemented to enhance the realism and dynamism of the emergency response scenarios.

A. Vehicle Impact Dynamics

As described in Section VII, the SUV rigid-body simulation constitutes the primary physics component. The vehicle's mass was set to 1800 kg, restitution to 0.3, and friction to 0.8, accurately representing automotive physics. Contact shadows and motion blur (shutter angle 180°) were enabled to reinforce the sense of violent impact, with the motion blur strength tuned to 0.5 for optimal visual effect.

B. Atmospheric Weather Particles

A particle system emitting 50,000 small mesh instances (elongated icospheres, scale 0.02) was attached to a large invisible plane above the accident scene to simulate rain. Particles follow a $-Y$ world-space force field with randomised turbulence (strength 1.5) to create realistic angled rainfall with variation. A secondary particle system emits 3,000 spark/debris objects from the collision impact point using an initial burst with high velocity and drag, representing flying debris and glass fragments. These particle systems contribute significantly to the atmospheric quality of the accident scene.



Fig. 15: Aerial night view: particle-based rain and debris visible around the accident scene with ambulance responding in background.

IX. AI TOOLS USAGE

In compliance with the course AI usage policy, the following AI tools were employed:

- **ChatGPT 4o** – used to generate initial Python-scripted Blender driver expressions for the ambulance light-bar flicker animation, and for documentation assistance in structuring this report. The generated scripts were reviewed and modified by team members to ensure correctness.
- **Midjourney v6** – used as a reference mood board for scene colour grading and lighting direction; no AI-generated image was used directly in the rendered output. The mood boards provided inspiration for cinematic composition and atmospheric lighting.

All creative modelling, rigging, animation, and rendering work was performed manually by team members in Blender, ensuring original artistic contribution.

X. ASSET CREDITS

External assets used in the project are listed in Table I. These assets were selected to supplement the team's original work while maintaining consistent visual quality.

TABLE I: External assets and their sources.

Asset	Source	Licence
City building pack	Sketchfab	CC-BY 4.0
Street furniture	BlenderKit	Free tier
Character base meshes	Mixamo	Adobe EULA
Ambulance textures	Textures.com	Standard
Tree foliage	Botaniq (lite)	GPL

XI. MEMBER CONTRIBUTIONS

TABLE II: Team member contribution summary.

Member	Contributions
Member (Name)	1 Virtual environment modelling (city block, roads, buildings); lighting setup for outdoor scenes.
Member (Name)	2 Hero object – ambulance vehicle: modelling, UV unwrapping, texture and material creation.
Member (Name)	3 Hero object – stretcher; character rigging (paramedic and patient); stretcher-push animation.
Member (Name)	4 Hospital interior modelling (hero object – reception desk); character animation (receptionist); fire/alarm lighting.
Member (Name)	5 Physics simulations (rigid-body collision, rain particle system); Cycles rendering, compositing, and video editing.

XII. CONCLUSION

This project successfully demonstrates a comprehensive virtual ambulance emergency response simulation produced entirely within Blender. Three distinct scenarios – a night-time traffic collision, a patient stretcher evacuation, and a hospital fire emergency – were animated and rendered with photorealistic lighting, skeletal character motion, and physics-based effects. Key contributions include the three original hero objects (ambulance, stretcher, and reception desk), a fully rigged humanoid character pipeline based on the Mixamo rig, and a multi-layered lighting system that transitions from warm street-lamp ambience to dramatic red emergency floods. The integration of rigid-body physics for vehicle dynamics and particle systems for atmospheric effects adds significant realism to the simulation. The simulation meets all seven required project components and provides a solid foundation for future extension into an interactive training experience using a game engine such as Unreal Engine or Godot. Potential enhancements include interactive user control, patient vital signs monitoring, and branching scenario paths for training purposes.

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